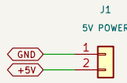
1. POWER – external regulated 5V / XIAO onboard 3.3V regulator

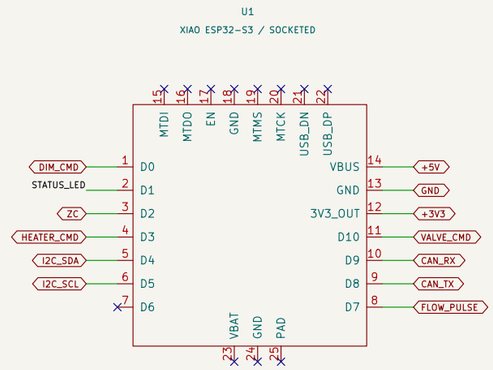


J1 feeds XIAO 5V and the existing 5V interfaces.
XIAO 3V3 supplies CAN VIO, I2C, TMP102 and FLOW pull-up.
USB-C, LDO, BOOT and RESET are on the removable XIAO.
Disconnect external J1 power before USB flashing.



3V3 PWR_FLAG represents the regulator output on the XIAO.

2. XIAO ESP32-S3 – two female 1x7 sockets, 2.54mm pitch

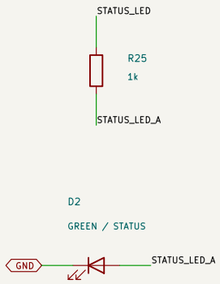


U1: D3/GPIO4 BOILER; D1/GPIO2 LED; D4/GPIO5 SDA; D5/GPIO6 SCL.
D7/GPIO44 FLOW; D8/GPIO7 CAN TX; D9/GPIO8 CAN RX; D10/GPIO9 VALVE.

Seeed numbering: 1..7 = D0..D6; 8..11 = D7..D10.
12=3V3 OUT; 13=GND; 14=5V IN. Pins 15..25 are bottom pads, NC.
D0/GPIO1 DIM_CMD output; D2/GPIO3 ZC input; D6/GPIO43 unused.

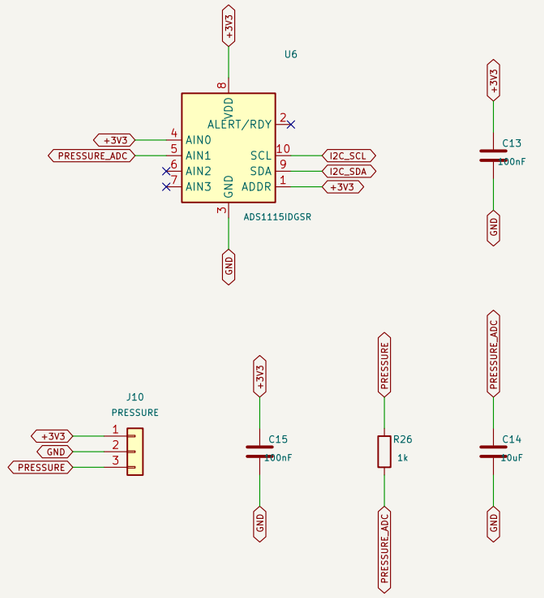
Footprint: Sensors_Local:XIAO-ESP32-S3-DIP-Headers.
Fit two female 1x7 sockets; XIAO fitted manually from stock.
Row spacing 15.24mm; 2.54mm pitch. GPIO4 avoids boot straps.

3. STATUS LED – GPIO2 / D1, HIGH = ON



Existing green LED and 1k resistor retained.

10. PRESSURE ADC – optional ADS1115, J10 = screen_sensors J3 order



U6 ADS1115 on the shared I2C bus, ADDR tied to 3V3: 0x49 (TMP102 already uses 0x48).
A0 reads the local 3V3; A1 reads pressure through R26/C14 (1k / 10uF, 16 Hz, 10 ms).
A2, A3 and ALERT/RDY unused. J10 pin order matches screen_sensors J3: same cable.
XDB401 output 0.4–2.4V on 3V3 supply, not ratiometric: use the absolute A1 voltage.

FUTURE PCB FIRMWARE: BOILER GPIO4, active HIGH (MOSFET driver).
Preload LOW before output enable; disable heater pull-up.
Current HW-399 firmware uses GPIO3, active LOW: do not use unchanged.
Other existing XIAO pins retained; add status LED on GPIO2.
Firmware unchanged; PCB placement and routing pending.

Sensors interfaces

File: sensors_interfaces.kicad_sch

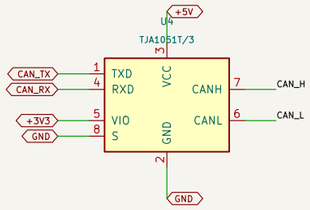
Socketed XIAO ESP32-S3 / Seeed DIP footprint
CoffeeFlow

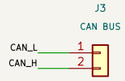
Sheet: /
File: sensors.kicad_sch

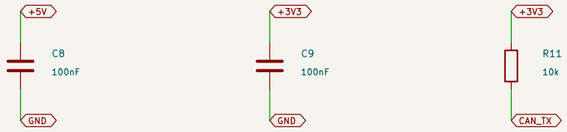
Title: Sensors – XIAO ESP32-S3 sockets and status LED

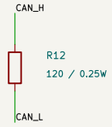
Size: A3	Date: 2026-10-07	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/2

4. CAN — TJA1051T/3, 5V bus driver / 3.3V logic



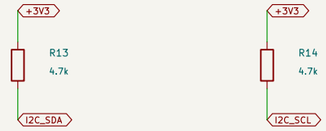


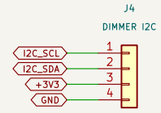


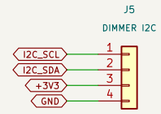


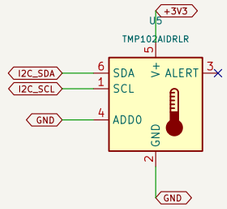
S/SLNT hard-wired low. R11 keeps TX recessive during reset.
R12 permanently terminates this bus-end node (120 ohms).
J3 has no GND pin: power GND must run alongside CAN pair.

5. SHARED I2C — 3.3V, Grove/HY2.0 cable order





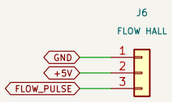




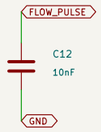


Grove opening up: left-to-right GND / 3V3 / SDA / SCL.
Numbered pads: 1=SCL yellow, 2=SDA white, 3=3V3 red, 4=GND black.
Addresses: dimmer 0x50 / XDB401 0x7F / TMP102 0x48.
TMP102 ADD0 grounded; ALERT unused. Mount away from XIAO and MOSFET heat.
Pull-ups always fitted: 2.35k effective with XDB401 attached.
100kHz initial bus speed; remote modules provide local supply decoupling.

6. DIGMESA — open collector, existing RC filter

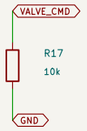


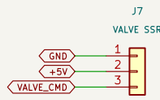




GPIO44 / D7 input; internal pull-up disabled. 1k x 10nF = 10us.
Digmesa powered from 5V; signal pulled to 3.3V only.
Pin order: GND brown / 5V green / SIGNAL yellow.
RC discharge speed also depends on sensor/cable impedance.

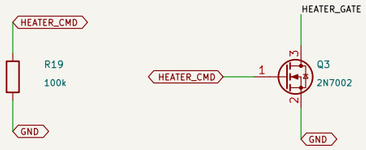
7. VALVE SSR — 3.3V signal / 5V supply



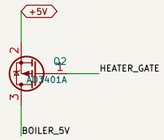


GPIO9 / D10, 3.3V signal / 5V VCC; default LOW.

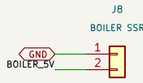
8. BOILER SSR — powered 5V output, HIGH command = ON







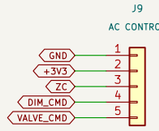




J8: SSR (-) -> pin1 GND; SSR (+) -> pin2 switched 5V.
Q3 pulls Q2 gate LOW; Q2 supplies the SSR current.
GPIO4 / D3 -> HEATER_CMD (HIGH = ON).
R19/R21/R22 ensure off with GPIO floating or 3V3 absent. Shared GND.
Gate pull-up current about 0.5mA when ON; load <=100mA; no mains.

9. AC CONTROL — optional accontrol board, same XH order

J9 = accontrol J1: 1 GND, 2 3V3, 3 ZC, 4 DIM, 5 SSR.
ZC: GPIO3 / D2 input, 10k pull-up on accontrol; enable internal pull-down.
DIM_CMD: GPIO1 / D0 output, 4.7k pull-down on accontrol (low at boot).
VALVE_CMD shared with J7: fit one valve driver only. 3V3 load <=25mA.



XIAO sockets — schematic only; PCB pending

CoffeeFlow

Sheet: /Sensors interfaces/
File: sensors_interfaces.kicad_sch

Title: Sensors — CAN, I2C, flowmeter and SSR outputs

Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/2